

114-3海攬班高二下重補修_測驗

Topic 1: 2D Newton's Laws & Forces



Useful Formulas for kinematics

$$(1) v = v_0 + at$$

$$(2) \Delta x = v_0 t + \frac{1}{2}at^2$$

$$(3) v^2 = v_0^2 + 2a\Delta x$$

Q1

A 20 kg table sits on a flat floor. The static friction coefficient is $\mu_s = 0.6$ and the kinetic friction coefficient is $\mu_k = 0.4$.

- (a) Calculate the normal force acting on the table.
- (b) You push the table with 100 N of force. Does the table move? What is the friction force pushing back against you right now?

Q2

Based on Q1, now you push harder with 150 N of force.

- (a) Which friction (static or kinetic) acts on it now?
- (b) Focus on the X-direction. Calculate the net force.
- (c) Calculate the acceleration of the table.

Q3

Two dogs are pulling a toy bone. Dog A pulls West with 60 N. Dog B pulls South with 80 N.

- (a) Draw a dot representing the toy bone, and draw these two force vectors acting on it.
- (b) Use the Pythagorean theorem to calculate the magnitude of the net force on the toy.

Q4

You are pulling a 50 kg sled on flat snow. Your pulling force is 100 N at an angle of 37° above the horizontal. The kinetic friction coefficient is $\mu_k = 0.1$.

- (a) Calculate the horizontal pull (F_x) and the vertical pull (F_y).

- (b) Calculate the normal force from the snow.
- (c) Find the net force in the X-direction.

Q5

A 50 kg skier is sliding down a ski slope tilted at 53° . The kinetic friction coefficient between the skis and the snow is $\mu_k = 0.5$.

- (a) Calculate the force parallel to the slope and the force perpendicular to the slope respectively.
- (b) What is the value of the normal force? Calculate the kinetic friction force.
- (c) Calculate the acceleration of the skier sliding down the slope.

Q6

Imagine a 20kg lawnmower.

- Situation A:

You **pull** with a force of 100 N at an angle 37° *above* the horizontal.

- Situation B:

You **push** with a force of 100 N at an angle 37° *below* the horizontal.

- (a) Draw the free-body-diagram (FBD) for both A and B situation. In which situation will the normal force be larger? Explain your reasoning based on the FBD.
- (b) Which situation (Pushing or Pulling) will have a larger friction force pushing back against you? Explain the reason.
- (c) Calculate the exact normal force for Situation B.

Topic 2: Circular Motion

Q1

Imagine a flat playground spinner. The distance between the center and the edge of the spinner is $r = 3$ m. It is a perfectly flat, solid circular platform that spins horizontally.

- (a) The spinner rotates through an angle of $\theta = 4$ rad. Calculate the actual distance you traveled if you sit at the edge.
- (b) If it takes $t = 2$ s to complete this rotation, calculate the angular velocity (ω) of the spinner.
- (c) Calculate your linear speed.



Q2

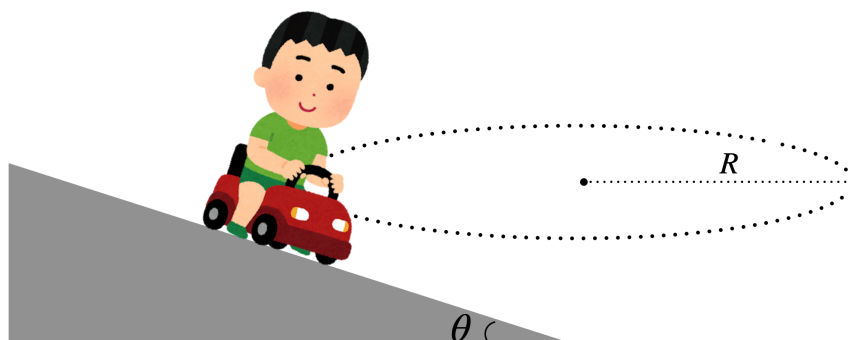
Imagine you place a small coin on the very edge of the same spinner from Q1.

- Calculate the centripetal acceleration of the coin while it is turning on the spinner.
- Which direction does this centripetal acceleration point? Depict circle as the top view of the spinner, and denote the acceleration arrow.
- Suddenly, the coin slips! Which direction does the coin fly in the horizontal direction? Explain using Newton's 1st Law.

Q3

Imagine a 1200 kg car turning on a smooth, frictionless road that is banked (tilted) at an angle of $\theta = 37^\circ$. The car is making a HORIZONTAL circle.

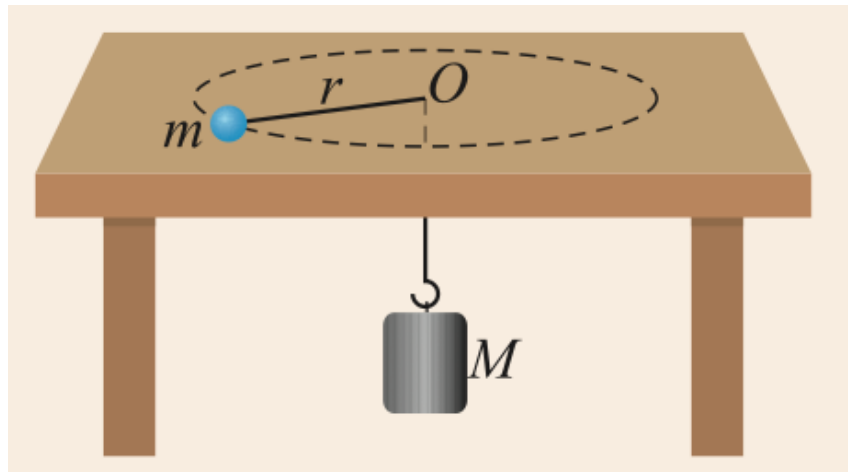
- Calculate the exact value of the normal force.
- Calculate the centripetal force (F_c).
- Calculate the centripetal acceleration (a_c) of the car.



Q4

A small blue ball with mass $m = 5 \text{ kg}$ is on a frictionless horizontal table. It is tied to a string that passes through a hole in the center to a hanging weight with mass $M = 4 \text{ kg}$. The blue ball is swinging in a horizontal circle with a radius of $r = 2 \text{ m}$.

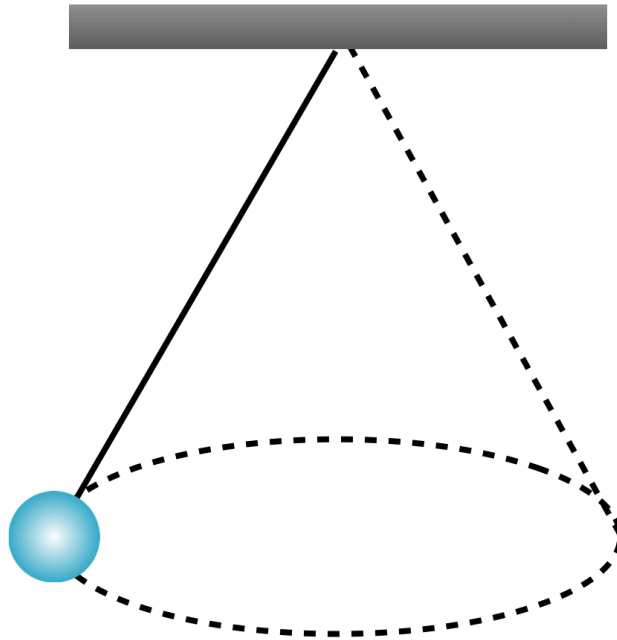
- (a) Calculate the Tension in the string.
- (b) Calculate the centripetal acceleration (a_c) of the blue ball.
- (c) Calculate the blue ball's tangential speed.



Q5

A ball with a mass of $m = 1.2 \text{ kg}$ is flying in a horizontal circle, suspended by a string of length $L = 1.5 \text{ m}$. The string makes an angle of $\theta = 53^\circ$ with the vertical straight line.

- (a) Calculate the exact value of the tension.
- (b) Calculate the radius R of the horizontal circle and calculate the centripetal force.
- (c) Calculate the tangential speed of the ball.



Q6

Analyze the Conical Pendulum further. Imagine you want the same toy plane to fly in a **WIDER circle** (meaning the angle θ becomes larger and the plane is higher up).

(a) Even if the circle is wider, the plane is still making a horizontal circle without falling. Does the vertical upward force (T_y) need to change? Explain your reason. (Hint: Look at Gravity).

(b) Because the angle θ is larger, the string is more "flat". To keep the SAME vertical force T_y while pulling at a flatter angle, must the total tension (T) in the string increase or decrease? Explain your reason.

(c) To maintain this new, wider circular path, does the plane need to fly faster or slower? Explain your reason.